GNSS-Independent Uplink LEO Satellite Synchronization

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***Abstract*— In a nonterrestrial network (NTN) with low earth orbit (LEO) satellites, there could be substantial Doppler shifts due to the relative movement between the satellites and the user equipment (UE). If the UE does not have global navigation satellite system (GNSS) capability, pre-compensation of the Doppler shifts cannot be made. Thus, synchronization signals are required which could be detected subject to large frequency offsets. We propose superposed pairs of interleaved Zadoff-Chu (SPI-ZC) synchronization sequences, each ZC sequence being interleaved by a linear permutation polynomial (LPP). The LPP renders a large set of sequences and sequences with improved peak-to-sidelobe-ratio (PSR) in the periodic correlation function, which improves the probability of correct synchronization. Lower detection complexity can be achieved compared to prior synchronization methods.**

***Index Terms*— Satellite, synchronization, Zadoff-Chu sequence.**

I. Introduction

THE 3GPP New Radio (NR) 5G system has recently been integrated with nonterrestrial networks (NTNs) supporting satellite communications, as a means to improve network coverage. This includes, e.g., low earth orbit (LEO) satellites located at a heights of 300-1500 km [1]. The movement of the satellite causes Doppler shift of the received signal, e.g., for a LEO satellite at 600 km height, a frequency offset of up to 48 kHz is assumed [2], which may be larger than the subcarrier spacing of the orthogonal frequency division multiplexing (OFDM) signal. Even with such large frequency offset, the LEO satellite must be able to determine timing and frequency synchronization from a signal, e.g., a random access preamble, transmitted by the user equipment (UE). 3GPP assumed that all NTN-capable UEs are equipped with a global navigation satellite system (GNSS) unit and that the satellite ephemeris can be signaled to the UE. That makes it possible for the UE to determine the position and velocity of itself and the satellite, and thus to compute the frequency offset, which is used for pre-compensation of the Doppler shift of the transmitted signal from the UE. Thereby, the issue with large Doppler shift of the received signal at the LEO satellite is eliminated. However, GNSS capability may be reconsidered, since the GNSS signal could be weak and/or low-cost UEs might neither have the time nor the energy to determine its position or receive the ephemeris transmissions. In particular, GNSS-independent operation for NTN has been listed as an enhancement in NR Release-19 [3]. Hence, it is a relevant research problem to design synchronization signals which are robust to large frequency offsets in this context.

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Synchronization subject to large frequency offsets may require new detection methods [4], [5] or new waveforms and sequences. A dual-chirp waveform was considered in [6] and [7] for synchronization in satellite communications, which is a superposition of one up-chirp and one down-chirp signal. In terms of sequences, considered approach was to transmit pairs of Zadoff-Chu (ZC) sequences, either concatenated [8], or superposed [9], [10]. A dual ZC sequence, as a superposition of a ZC sequence and its complex conjugate version was proposed in [11]. A generalization, where the complex conjugated version is cyclically shifted and multiplied with a complex constant, was proposed in [12] and [13]. Superposed ZC sequences from different root indices were proposed in [14].

The synchronization performance of superposed ZC sequences depends on the root indices of the constituent ZC sequences. Therefore, the number of usable superposed ZC sequences may become limited, considering that some of them perform poorly. The issue with poor performance of certain root indices has not been well addressed in prior work. In this letter, we propose synchronization signal based on superposed pairs of interleaved ZC (SPI-ZC) sequences. The interleaver serves to increase the peak-to-sidelobe-ratio (PSR) of a periodic correlation function, which improves the probability of correct timing and frequency synchronization. Furthermore, by constructing SPI-ZC sequences from different interleavers, a large set of sequences can be produced. This may be required if the sequences are used as random access preambles, where many unique preambles are needed per cell, as well as having different sets of preambles in different cells.

II. Uplink Synchronization

A. Synchronization Signal

Let $k = 0, 1, \ldots, N-1$, and define the ZC sequence

$$x_u[k] = e^{-\mathrm{j}\frac{\pi}{N}uk(k+1)} \tag{1}$$

where N is an odd integer, $\mathrm{j} = \sqrt{-1}$, u is a positive integer, aka. root index, for which $\gcd(u, N) = 1$, where $\gcd(v, w)$ denotes the greatest common divisor of v and w.

Consider the linear permutation polynomials (LPPs) $g^{(0)}[k] = g_a k + g_b \pmod N$ and $g^{(1)}[k] = g_c k + g_d \pmod N$ where $g_a, g_c \in \{1, 2, \ldots, N-1\}$ and $g_b, g_d \in \{0, 1, \ldots, N-1\}$ with $\gcd(g_a, N) = \gcd(g_c, N) = 1$ and $\pmod N$ is the modulo-N operator. We propose the following SPI-ZC sequence for timing and frequency synchronization on the uplink (i.e., the feeder link), e.g., as a random access preamble

$$y[k] = x_{u_0}[g^{(0)}[k]] + x_{u_1}[g^{(1)}[k]] \tag{2}$$

where $g^{(i)}[k]$ interleaves the ith ZC sequence [15]. Furthermore, we require that

$$\gcd(u_0 g_a^2 - u_1 g_c^2, N) = 1 \tag{3}$$

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implying that the inverse $(u_0 g_a^2 - u_1 g_c^2)^{-1}$ exists modulo-N, which is utilized in the synchronization method, as will be shown in the next subsection.

The transmitted signal is obtained from the N-point discrete Fourier transform (DFT) of (2) where the DFT coefficients, $Y[n], n = 0, \ldots, N-1$, are mapped to N consecutive subcarriers in an OFDM signal with subcarrier spacing Δf. The time-discrete signal $\tilde{y}[l], l = 0, \ldots, M-1$ is obtained from an inverse DFT (IDFT) of size M on $Y[n]$. A cyclic prefix of N_{CP} samples is included by letting $l = -N_{\mathrm{CP}}, -N_{\mathrm{CP}} + 1, \ldots, M-1$. In the subsequent analysis, we will assume $M = N$, for which $\tilde{y}[k] = y[k]$. We also assume that the sampling rate is $f_s = N\Delta f$ in both transmitter and receiver.

B. Timing and Frequency Synchronization

We assume an additive white Gaussian noise (AWGN) channel where the base-band received signal, $r[k]$, is subject to frequency offset f and a delay of t samples, i.e.,

$$r[k] = y[k-t]e^{\mathrm{j}\frac{2\pi}{N}fk} + n[k] \tag{4}$$

wherein $n[k]$ is AWGN. The normalized frequency offset $f = f_D/\Delta f$ includes the Doppler shift, f_D, and comprises an integer part, $f_{\mathrm{int}} \in \{0, 1, \ldots, H-1\}$, where H is an odd integer, and a fractional part, $f_{\mathrm{frac}} \in [-0.5, 0.5]$, such that

$$f = f_{\mathrm{int}} - (H-1)/2 + f_{\mathrm{frac}} \tag{5}$$

where the term $(H-1)/2$ is introduced to allow both positive and negative integer offsets. The value of H is known to the receiver since it is a parameter related to the maximum frequency offset $f_D^{\max}$ of the LEO satellite system, as

$$H = 2\lfloor f_D^{\max}/\Delta f \rfloor + 1. \tag{6}$$

where $\lfloor \cdot \rfloor$ is the floor operator.

The receiver should estimate t and f_{int}, subject to the uncompensated fractional frequency offset f_{frac}. Estimation of f_{frac} may be carried out subsequently, after t and f_{int} are determined, e.g., by iterative methods [11], from crosscorrelation with reference signals, or by using autocorrelation-based method on repetitive OFDM symbols parts [16]. The receiver determines two periodic correlation functions for $i = 0, 1$ and $d = 0, 1, \ldots, N-1$ according to

$$\tilde{\rho}_i(d) = \sum_{k=0}^{N-1} r[k] x_{u_i}^*[g^{(i)}[k-d]] \tag{7}$$

where $(.)^*$ denotes complex conjugate. Utilizing (2) and (7), the correlation functions are obtained as in (8) and (9), shown at the bottom of the page, where $\tilde{n}_i[k]$ is the output of correlator i in (7) with $n[k]$ as input and

$$\alpha_0 = e^{\mathrm{j}\frac{\pi}{N}u_0(g_a(t-d)+g_a^2(d^2-t^2)+2g_a g_b(t-d))} \tag{10}$$

$$\alpha_1 = e^{\mathrm{j}\frac{\pi}{N}u_1(g_c(t-d)+g_c^2(d^2-t^2)+2g_c g_d(t-d))} \tag{11}$$

$$\beta_0 = e^{\mathrm{j}\frac{\pi}{N}u_0(-g_a d+g_a^2 d^2-2g_a g_b d+g_b^2+g_b)} e^{-\mathrm{j}\frac{\pi}{N}u_1 g_c(-t+g_c t^2)} \times e^{-\mathrm{j}\frac{\pi}{N}u_1(g_d^2+g_d-2g_d g_c t)} \tag{12}$$

$$\beta_1 = e^{\mathrm{j}\frac{\pi}{N}u_1(-g_c d+g_c^2 d^2-2g_c g_d d+g_d^2+g_d)} e^{-\mathrm{j}\frac{\pi}{N}u_0 g_a(-t+g_a t^2)} \times e^{-\mathrm{j}\frac{\pi}{N}u_0(g_b^2+g_b-2g_b g_a t)}. \tag{13}$$

By evaluating the geometric sum in (8), we have

$$|\rho_{00}(d)| = \left| \frac{\sin(\pi(u_0 g_a^2(t-d)+f))}{\sin\left(\frac{\pi(u_0 g_a^2(t-d)+f)}{N}\right)} \right|. \tag{14}$$

The effect of f is that it shifts the location of the maximum value of (14). To derive the estimates of t and f_{int}, we will let $f_{\mathrm{frac}} = 0$, in which case we have

$$|\rho_{00}(d)| = \begin{cases} N, & u_0 g_a^2(t-d) + f_{\mathrm{int}} - (H-1)/2 \equiv 0 \,(\mathrm{mod}\, N) \\ 0, & \text{otherwise.} \end{cases} \tag{15}$$

If $f_{\mathrm{frac}} \neq 0$, then $|\rho_{00}(d)| < N$ since the argument in the numerator's sine function (14) will not be a multiple of π. By performing the same expansion for $\rho_{11}(d)$ and defining $d_0 = \arg\max_d |\tilde{\rho}_0(d)|$ and $d_1 = \arg\max_d |\tilde{\rho}_1(d)|$, t and f_{int} can be estimated from a linear congruence equation system obtained from $|\rho_{00}(d)|$ and $|\rho_{11}(d)|$ as

$$u_0 g_a^2(\hat{t} - d_0) + \hat{f}_{\mathrm{int}} - (H-1)/2 \equiv 0 \,(\mathrm{mod}\, N) \tag{16}$$

$$u_1 g_c^2(\hat{t} - d_1) + \hat{f}_{\mathrm{int}} - (H-1)/2 \equiv 0 \,(\mathrm{mod}\, N) \tag{17}$$

which has the solutions

$$\hat{t} = (u_0 g_a^2 - u_1 g_c^2)^{-1}(u_0 g_a^2 d_0 - u_1 g_c^2 d_1) \,(\mathrm{mod}\, N) \tag{18}$$

$$\hat{f}_{\mathrm{int}} = u_0 g_a^2 d_0 - u_0 g_a^2 t + (H-1)/2 \,(\mathrm{mod}\, N). \tag{19}$$

III. SPI-ZC Sequence Design

Let $\rho_0(d) = \rho_{00}(d) + \rho_{01}(d)$ and $\rho_1(d) = \rho_{11}(d) + \rho_{10}(d)$. The probability of $\hat{t} = t$ and $\hat{f}_{\mathrm{int}} = f_{\mathrm{int}}$ depends on the magnitude of the main peak and the sidelobes of $|\rho_0(d)|$ and $|\rho_1(d)|$. These can be shaped by the selection of u_0, u_1, g_a, g_b, g_c and g_d. Preferably, $|\rho_i(d_i)|$ should be large

$$\tilde{\rho}_0(d) = \underbrace{\alpha_0 \sum_{k=0}^{N-1} e^{\mathrm{j}\frac{2\pi}{N}(u_0 g_a^2(t-d)+f)k}}_{\rho_{00}(d)} + \underbrace{\beta_0 \sum_{k=0}^{N-1} e^{\mathrm{j}\frac{\pi}{N}u_0(g_a^2 k^2+(g_a-2g_a^2 d+2g_a g_b)k)} e^{-\mathrm{j}\frac{\pi}{N}u_1(g_c^2 k^2+(g_c-2g_c^2 t+2g_c g_d)k)} e^{\mathrm{j}\frac{2\pi}{N}f}}_{\rho_{01}(d)} + \tilde{n}_0[k] \tag{8}$$

$$\tilde{\rho}_1(d) = \underbrace{\alpha_1 \sum_{k=0}^{N-1} e^{\mathrm{j}\frac{2\pi}{N}(u_1 g_c^2(t-d)+f)k}}_{\rho_{11}(d)} + \underbrace{\beta_1 \sum_{k=0}^{N-1} e^{\mathrm{j}\frac{\pi}{N}u_1(g_c^2 k^2+(g_c-2g_c^2 d+2g_c g_d)k)} e^{-\mathrm{j}\frac{\pi}{N}u_0(g_a^2 k^2+(g_a-2g_a^2 t+2g_a g_b)k)} e^{\mathrm{j}\frac{2\pi}{N}f}}_{\rho_{10}(d)} + \tilde{n}_1[k] \tag{9}$$

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while $|\rho_i(d_i)|, d \neq d_i$ should be small. Since this should be achieved for any delay t and frequency offset f, the problem of determining sequence parameters could be considered without loss of generality assuming $t = 0$ and $f = 0$. Consider $i = 0$, then from (15), it follows that $-u_0 g_a^2 d_0 \equiv 0 \,(\mathrm{mod} N)$, when $d = d_0$. Therefore, from (8) we obtain (20), shown at the bottom of the page. A corresponding expression can be derived for $|\rho_1(d_1)|$. By Property 1 in Appendix, we have $|S| = \sqrt{N}$, so it follows from (20) that

$$N - \sqrt{N} \leq |\rho_i(d_i)| \leq N + \sqrt{N} \tag{21}$$

which implies that the $\mathrm{PSR}_i(d) = |\rho_i(d_i)|/|\rho_i(d)|, d \neq d_i$ is bounded by

$$\sqrt{N} - 1 \leq \mathrm{PSR}_i(d) \leq \sqrt{N} + 1 \tag{22}$$

because when $d \neq d_0$, (15) implies that $\rho_{00}(d) = 0$ and it follows from (20) and Property 1 that $|\rho_0(d)| = \sqrt{N}$ (it can also be shown that $|\rho_1(d)| = \sqrt{N}, d \neq d_1$). Therefore, the magnitude of the sidelobes is independent of d and the sequence parameters. Hence, the PSR is independent of d and the parameters which maximize $|\rho_i(d_i)|$ also maximize the PSR. The problem to maximize the PSR is thus equivalent to, for $i = 0$ or $i = 1$, solve

$$\underset{u_0, u_1, g_a, g_b, g_c, g_d}{\arg\max} \; |\rho_i(d_i)|. \tag{23}$$

Finding solutions to (23) can be simplified by utilizing closed-form expressions of S in (20), which are possible for certain combinations of sequence parameters, as follows. By defining $a = u_0 g_a^2 - u_1 g_c^2$ and $b = u_0(g_a + 2g_a g_b) - u_1(g_c + 2g_c g_d)$, and using $aN + b \equiv 0 \,(\mathrm{mod}\, 2)$, which follows from Lemma 1 in Appendix, it can be shown that

$$S = G(a, b, 2N)/2 \tag{24}$$

where $G(v, w, q) = \sum_{k=0}^{q-1} e^{\mathrm{j}2\pi(vk^2 + wk)/q}$ is the generalized quadratic Gauss sum. If b/a is an even integer and $c = \gcd(a, 2N)$, we can complete the square of the exponential function, remove the integer shift $b/2a$ from the terms and obtain

$$G(a, b, 2N) = e^{-\mathrm{j}\frac{\pi b^2}{4Na}} \sum_{k=0}^{2N-1} e^{\mathrm{j}\frac{2\pi}{2N/c}\frac{a}{c}\left(k + \frac{b}{2a}\right)^2} \tag{25}$$

$$= e^{-\mathrm{j}\frac{\pi b^2}{4Na}} c G(a/c, 0, 2N/c). \tag{26}$$

From Lemma 2 in Appendix, $c = 2$, and we can therefore use the closed-form expressions for $G(a/2, 0, N)$ from [17]. If b/a is an odd integer, we can proceed similarly, remove the integer shifts from the terms and obtain

$$G(a, b, 2N) = e^{-\mathrm{j}\frac{\pi b^2}{4Na}} \sum_{k=0}^{2N-1} e^{\mathrm{j}\frac{2\pi}{2N/c}\frac{a}{c}\left(k + \frac{1}{2}\right)^2} \tag{27}$$

$$= e^{-\mathrm{j}\frac{\pi b^2}{4Na}} c H(a/c, 2N/c) \tag{28}$$

where the sum $H(\cdot, \cdot)$ exists in closed-form [17, Ch. 1, Exercise 23]. Here, it is possible that either $c = 1$ or $c = 2$. The $\mathrm{PSR} = \sqrt{N} + 1$ according to (20) when

$$\gamma S = \sqrt{N}. \tag{29}$$

By using (24), (26) and (28), it can be checked whether (29) holds, i.e., if the sequence parameters achieve the maximum PSR, which could simplify the search for solutions to (23). Notably, these are sufficient conditions, since it is possible that the maximum PSR is achieved even if b/a is not an integer. The objective function (23) serves to determine the parameters of the transmitted random access preamble. As there might be many different preambles in the system, which possibly can be allocated and/or reallocated to different UEs, these preambles, and thus their defining parameters, have to be available before these preambles are deployed, e.g., in a standard specification. It means that the solutions of (23) can be determined during a system/standard design phase, and then be stored/specified in a compact table that can be used in both the transmitter and the receiver.

IV. Numerical Results

A. SPI-ZC Sequence Properties

The results of exhaustive evaluations are contained in Table I, which shows the maximum achievable PSR and the number of corresponding unique sequences for this PSR. A sequence is considered unique if it is not a cyclically shifted version of another sequence in the set and/or is not equal to another sequence multiplied with a constant. For comparison, the results without LPPs are also shown, i.e., when $g_a = g_c = 1$ and $g_b = g_d = 0$. This produces sequences from [14]. It can be concluded that SPI-ZC sequence can offer several order of magnitudes larger set of sequences and at the same time provide slightly larger PSR. The third column shows how many sequences there are where b/a is an integer and which achieve the maximum PSR. We observe that if $N \equiv 1 \,(\mathrm{mod}\, 4)$, there exist sequence parameters such that $\mathrm{PSR} = \sqrt{N} + 1$.

Table II gives the peak-to-average-power-ratio (PAPR), $\mathrm{PAPR} = \max_l |\tilde{y}[l]|^2 / \sum_{l=0}^{M-1} |\tilde{y}[l]|^2 / M$, with $M/N = 6$ times oversampling, for the set of sequences achieving the maximum PSR according to Table I. The set of SPI-ZC sequences is larger and there is a larger spread in the PAPRs; typically the minimum PAPR is smaller and the maximum PAPR is larger, while the median PAPRs are in many cases comparable. Table III contains the crosscorrelation (i.e., inner product) among the sequences in the set. Albeit the sequence set is much larger for SPI-ZC, the minimum and median crosscorrelation values are smaller. It should be noted that SPI-ZC sequences, in many cases, can produce orthogonal sequences.

$$|\rho_0(d_0)| = \Big| N + \underbrace{e^{\mathrm{j}\frac{\pi}{N}u_0(g_b^2 + g_b)} e^{-\mathrm{j}\frac{\pi}{N}u_1(g_d^2 + g_d)}}_{\gamma} \underbrace{\sum_{k=0}^{N-1} e^{\mathrm{j}\frac{\pi}{N}((u_0 g_a^2 - u_1 g_c^2)k^2 + (u_0(g_a + 2g_a g_b) - u_1(g_c - 2g_c g_d))k)}}_{S} \Big| \tag{20}$$

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TABLE I

MAXIMUM PSR AND THE CORRESPONDING NUMBER OF SEQUENCES OF LENGTH N, VALUES MARKED WITH † ARE WHERE $\mathrm{PSR} = 1 + \sqrt{N}$

	SPI-ZC			Seq. from [14]	
N	#seq.	#seq. from (26) and (28)	PSR	#seq.	PSR
5	13	3	3.2361†	3	3.1013
7	105	0	3.6275	5	3.6275
9	54	9	4.0000†	3	3.8205
11	495	0	4.3088	9	4.3088
13	411	64	4.6056†	11	4.5827
15	26	0	4.8686	3	4.8686
17	988	109	5.1231†	15	5.0685
19	2907	0	5.3561	17	5.3561
21	192	43	5.5826†	5	5.5734
23	5313	0	5.7939	21	5.7939
25	3250	188	6.0000†	15	5.9738

TABLE II

MINIMUM, MEDIAN AND MAXIMUM PAPR [dB] FOR THE SET OF SEQUENCES OF LENGTH N HAVING MAXIMUM PSR

	SPI-ZC			Seq. from [14]		
N	min	med.	max	min	med.	max
5	2.98	2.99	4.98	3.07	3.07	6.56
7	1.88	4.75	6.38	3.30	3.53	3.38
9	2.12	3.95	5.40	3.66	4.38	4.38
11	2.26	4.46	6.53	3.98	4.46	4.77
13	2.24	4.20	7.17	2.14	4.13	7.05
15	3.26	4.90	5.93	4.46	5.71	5.71
17	2.50	4.70	7.45	2.88	4.53	6.49
19	2.45	4.75	7.76	3.42	4.69	6.45
21	2.55	4.81	6.86	4.08	6.01	6.87
23	2.68	4.99	7.60	3.92	5.07	6.33
25	2.81	5.12	7.31	4.32	4.68	6.62

TABLE III

MINIMUM, MEDIAN AND MAXIMUM CROSSCORRELATION AMONG THE SEQUENCES OF LENGTH N IN THE SET HAVING MAXIMUM PSR

	SPI-ZC			Seq. from [14]		
N	min	med.	max	min	med.	max
5	0.0955	0.4434	0.7848	0.3809	0.5742	0.7676
7	0	0.3615	0.7748	0.3376	0.3857	0.6040
9	0.0434	0.3907	0.8197	0.5029	0.5029	0.5029
11	0	0.2638	0.7315	0.1992	0.2960	0.6524
13	0	0.2752	0.7171	0.0063	0.3277	0.7103
15	0	0.3096	0.7801	0.3565	0.5215	0.6866
17	0	0.2309	0.6952	0.0134	0.2963	0.6795
19	0	0.2056	0.6865	0.1713	0.2329	0.6132
21	0	0.2729	0.7416	0.2841	0.3580	0.5727
23	0	0.1787	0.6724	0.1608	0.2141	0.5825
25	0.0013	0.1790	0.7697	0.0952	0.3803	0.6597

B. Detection Probability

According to Table I, the maximum difference in PSR between sequences exists for $N = 9$ and we therefore select this value. Assuming $\Delta f = 15$ kHz and $f_D^{\max} = 48$ kHz, we obtain $H = 7$ from (6). A discrete uniform distribution is used for f_{int}. A uniform distribution is assumed for f_{frac}, which is not estimated or compensated for in the receiver. Without LPP, (20) implies that the PSR will only depend on $\Delta u = u_0 - u_1$ which has to fulfill $\gcd(\Delta u, N) = 1$ due to (3). The results of Monte-Carlo simulations in Fig. 1 confirm that the largest detection probability is achieved with SPI-ZC sequence because of the larger PSR ($\mathrm{PSR} = 4.0000$), compared to sequences from [14] with $\Delta u = 1\,(\mathrm{PSR} = 3.8205)$, $\Delta u = 4, \Delta u = 5$ ($\mathrm{PSR} = 2.0885$) and $\Delta u = 2, \Delta u = 7\,(\mathrm{PSR} = 3.3229)$. For example, about 8 dB signal-to-noise-ratio (SNR) gain of SPI-ZC sequences is achieved at

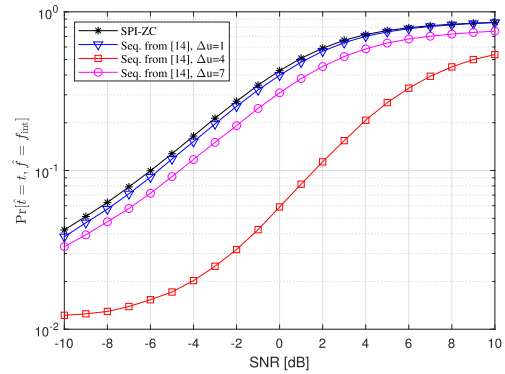

Fig. 1. Probability of detection as function of SNR for $N = 9$, for SPI-ZC sequence ($u_0 = 1, g_a = 4, g_b = 0; u_1 = 2; g_c = 2, g_d = 4$) and for the sequence from [14], for different values of $\Delta u = u_0 - u_1$.

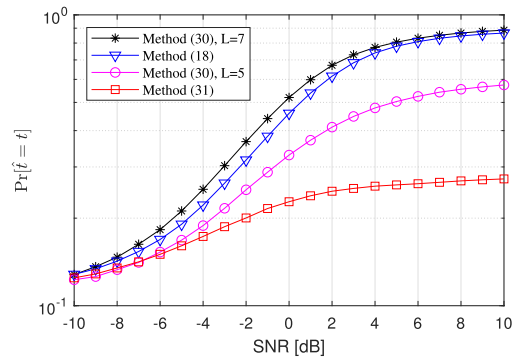

Fig. 2. Probability of timing detection as function of SNR for different detection methods of SPI-ZC sequence ($u_0 = 1, g_a = 4, g_b = 0$; $u_1 = 2; g_c = 2, g_d = 4$); method (18), method (30) with $L = 5$ or $L = 7$ and method (31).

10% detection probability, compared to sequences from [14] with $\Delta u = 4$. From Fig. 1 it can be observed that when $\Delta u = 1$, the sequences from [14] perform only slightly worse than SPI-ZC sequences. However, it should be kept in mind that there are much more SPI-ZC sequences than sequences from [14] that perform well, as shown in Table I.

C. Timing Detection Vs Receiver Complexity

The SPI-ZC preambles allow timing detection given by (18), requiring only two correlators in the receiver. A question is if the timing detection probability could be improved by increasing the receiver complexity. In Fig. 2, we compare (18) with two other timing detection methods, both using the same SPI-ZC sequence (2), but not utilizing the specific properties of the sequence. One method uses pre-compensation, where the received signal at each of L correlators is multiplied with a cancellation frequency, $f_c, c = 0, 1, \ldots, L-1$.

$$\rho(d, f_c) = \sum_{k=0}^{N-1} r[k] e^{-\mathrm{j}\frac{2\pi f_c k}{\Delta f N}} y^*[k - d \,(\mathrm{mod}\, N)] \tag{30}$$

The timing is determined by $\hat{d} = \arg\max_{d,c} |\rho(d, f_c)|$. We found that the largest detection probability is achieved when $L = H$ and $f_c/\Delta f = f_{\mathrm{int}} - (H-1)/2$, i.e., (30) becomes a bank of correlators matched to the integer frequency offsets. Using $L > H$ just increases the number of false timing candidates and showed no gain in detection probability. So $f_c/\Delta f \in \{0, \pm 1, \pm 2, \pm 3\}$ when $L = 7$ and we also include $L = 5$ with $f_c/\Delta f \in \{0, \pm 1.4, \pm 2.8\}$. The other

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$$|S|^2 = \sum_{k=0}^{N-1}\sum_{p=0}^{N-1} e^{\mathrm{j}\frac{\pi}{N}((u_0 g_a^2 - u_1 g_c^2)(k^2-p^2)+(u_0(g_a+2g_a g_b)-u_1(g_c+2g_c g_d))(k-p))} \tag{32}$$

$$= \sum_{k=0}^{N-1}\sum_{t=0}^{N-1} e^{\mathrm{j}\frac{\pi}{N}((u_0 g_a^2 - u_1 g_c^2)t(2k-t)+(u_0(g_a+2g_a g_b)-u_1(g_c+2g_c g_d))t)} \tag{33}$$

$$= \sum_{t=0}^{N-1} e^{\mathrm{j}\frac{\pi}{N}(-(u_0 g_a^2 - u_1 g_c^2)t^2+(u_0(g_a+2g_a g_b)-u_1(g_c+2g_c g_d))t)} \sum_{k=0}^{N-1} e^{\mathrm{j}\frac{2\pi}{N}(u_0 g_a^2 - u_1 g_c^2)tk} \tag{34}$$

method is performing correlation on two parts of the signal.

$$|\rho(d)| = \left|\sum_{k=0}^{(N-1)/2} r[k]y^*[k-d\,(\mathrm{mod}\,N)]\right| + \left|\sum_{k=(N+1)/2}^{N-1} r[k]y^*[k-d\,(\mathrm{mod}\,N)]\right| \tag{31}$$

Fig. 2 shows that (31) is much worse than (18). We can also conclude that a significant increase of the number of correlators, matched to H pre-selected frequencies by (30), brings only a marginal gain compared to (18).

V. Conclusion

The proposed SPI-ZC sequences offer better detection performance in channels with large Doppler shifts and also produce much more sequences than other superposed ZC sequences from the literature.

Appendix

Lemma 1: $(u_0 g_a^2 - u_1 g_c^2)N + u_0 g_a - u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$.

Proof: If $u_0 g_a - u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$, the following cases are possible:

i. $u_0 g_a \equiv 0\,(\mathrm{mod}\,2)$ and $u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$: This implies that $u_0 g_a^2 \equiv 0\,(\mathrm{mod}\,2)$ and $u_1 g_c^2 \equiv 0\,(\mathrm{mod}\,2)$.

ii. $u_0 g_a \equiv 1\,(\mathrm{mod}\,2)$ and $u_1 g_c \equiv 1\,(\mathrm{mod}\,2)$: This implies that $u_0 g_a^2 \equiv 1\,(\mathrm{mod}\,2)$ and $u_1 g_c^2 \equiv 1\,(\mathrm{mod}\,2)$.

If $u_0 g_a - u_1 g_c \equiv 1\,(\mathrm{mod}\,2)$, the following cases are possible:

iii. $u_0 g_a \equiv 0\,(\mathrm{mod}\,2)$ and $u_1 g_c \equiv 1\,(\mathrm{mod}\,2)$: This implies that $u_0 g_a^2 \equiv 0\,(\mathrm{mod}\,2)$ and $u_1 g_c^2 \equiv 1\,(\mathrm{mod}\,2)$.

iv. $u_0 g_a \equiv 1\,(\mathrm{mod}\,2)$ and $u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$: This implies that $u_0 g_a^2 \equiv 1\,(\mathrm{mod}\,2)$ and $u_1 g_c^2 \equiv 0\,(\mathrm{mod}\,2)$.

Therefore, $(u_0 g_a^2 - u_1 g_c^2)N + u_0 g_a - u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$. ■

Property 1: If S is defined as in (20), then $|S|^2 = N$.

Proof: We obtain (32)–(34), shown at the top of the page, by the substitution $t = k - p$ in (33), which is possible since the terms have period N. That is, if $z[t] = e^{\mathrm{j}\frac{\pi}{N}((u_0 g_a^2 - u_1 g_c^2)t^2 + (u_0(g_a+2g_a g_b)-u_1(g_c-2g_c g_d))t)}$, we obtain $z[t+N] = z[t]e^{\mathrm{j}\pi((u_0 g_a^2 - u_1 g_c^2)N+(u_0(g_a+2g_a g_b)-u_1(g_c-2g_c g_d)))} = z[t]$ because $(u_0 g_a^2 - u_1 g_c^2)N + u_0(g_a+2g_a g_b) - u_1(g_c+2g_c g_d) \equiv 0\,(\mathrm{mod}\,2)$ due to Lemma 1.

Now, the inner geometric sum in (34) is equal to N when $(u_0 g_a^2 - u_1 g_c^2)t \equiv 0\,(\mathrm{mod}\,N)$ and is equal to 0 otherwise. Due to (3), $t = 0$ is the only solution where the inner geometric sum is non-zero. Hence, we have $|S|^2 = N$. ■

Lemma 2: If $b/2a$ is an integer, then $\gcd(a, 2N) = 2$.

Proof: Due to (3), we have $\gcd(a, N) = 1$, so it is sufficient to show that $a \equiv 0\,(\mathrm{mod}\,2)$. Suppose $b = k2a$ for an integer k, implying that we require $b \equiv 0\,(\mathrm{mod}\,2)$. Then from the definition of b, we must have $u_0 g_a - u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$. From case i. and ii. in the proof of Lemma 1, it follows that if $u_0 g_a - u_1 g_c \equiv 0\,(\mathrm{mod}\,2)$, then $u_0 g_a^2 - u_1 g_c^2 \equiv 0\,(\mathrm{mod}\,2)$. That is, $a \equiv 0\,(\mathrm{mod}\,2)$. ■

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